

Ruby (Yuhan) Qiu, EIT Candidate

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HIGHLIGHTS OF QUALIFICATIONS

- **Embedded Systems & Motor Control:** Designed and implemented real-time motor control algorithms for autonomous robots, achieving 5% improved speed regulation, variation tolerance, and reduced overshoot by 35% using PID control and ROS integration.
- **FPGA & Hardware Design:** Developed 16-channel FPGA signal processing pipelines for wearable ultrasound and SD-card host systems, achieving <5 ms latency, 20% improved signal quality, and reliable data streaming at ~2 MB/s across multiple platforms.
- **Data Analysis & Visualization:** Skilled in Power BI, Excel, and MATLAB for processing energy, telematics, and biomedical data; prepared dashboards, KPIs, and reports supporting cross-functional decision-making and program optimization.
- **Cross-Functional Collaboration & Research:** Experienced in interdisciplinary collaboration with engineers, researchers, and clinicians; contributed to AI data annotation, embedded systems projects, and reliability analysis, ensuring compliance with ethics, standards, and regulatory requirements.

EDUCATION

Bachelor of Applied Science in Electrical Engineering

University of British Columbia

Sep 2022 - Jan 2027

Vancouver, Canada

EXPERIENCE

Research Assistant

Mar 2026 – Sep 2026

The Attentional Neuroscience Lab, UBC Department of Psychology

Vancouver, Canada

- **Software Development:** Developed a Python-based pipeline using the Azure Kinect Body Tracking SDK to process depth video data and extract gait parameters; enabled automated and scalable biomechanical analysis across multiple datasets.
- **Workflow Development & SOP Design:** Developed standardized workflows for gait data processing, including SDK setup, offline processing, and automated JSON/CSV data pipelines, improving reproducibility and efficiency of analysis.
- **Data Quality Standardization:** Established filming protocols (camera positioning, lighting, subject clothing, movement patterns) to ensure consistent and high-quality gait data, reducing noisy/invalid frames by ~30% and improving reliability of extracted parameters.
- **Automation & Pipeline Optimization:** Automated data processing steps, reducing manual intervention and accelerating batch processing efficiency by ~60%.

Navigation & Embedded Systems Member

Sep 2025 – May 2026

UBC Agrobot

Vancouver, Canada

- **Motor Control Design:** Designed and tested real-time motor control logic for an autonomous field robot, improving speed regulation accuracy by 5% and maintaining stable operation under load variations of up to $\pm 30\%$.
- **ROS Integration:** Implemented ROS and micro-ROS interfaces enabling reliable motor command and feedback exchange at 100 Hz, reducing communication latency by ~40% compared to the initial prototype.
- **Hardware Interfacing:** Integrated motor drivers, quadrature encoders, and motor controllers to achieve closed-loop control with encoder resolution up to 1024 PPR and position error below 2%.
- **Control Algorithms:** Developed and tuned PID-based motor control algorithms, decreasing overshoot by 35% and settling time by 20% during start-stop and turning maneuvers.
- **Embedded Programming:** Implemented motor control and communication modules in C++ and Python, supporting real-time execution on embedded hardware with over 99% uptime during field and bench testing.

AI Research Assistant

Jun 2025 - Nov 2025

Uterine Health Research Lab, UBC Faculty of Medicine

Vancouver, Canada

- **Medical Data Annotation:** Annotated and validated 250+ ultrasound imaging datasets for AI model development, achieving 100% compliance with de-identification, ethics approval, and licensing requirements.
- **Data Quality Assessment:** Evaluated dataset quality by reviewing 50+ medical publications and associated metadata, improving dataset consistency and usability for model training.
- **Research Documentation:** Documented dataset attributes, limitations, and quality metrics in a structured spreadsheet used by a 6-person research team for downstream development.
- **Cross-Functional Collaboration:** Collaborated with clinicians and AI researchers to define data selection and quality criteria, directly supporting successful preparation of datasets for model prototyping and validation.

- **Embedded Systems:** Assisted in the development and testing of vehicle telematics hardware and microcontroller-based systems, supporting 10+ GPS tracking devices and 5 CAN bus modules for fleet applications.
- **Testing & Validation:** Performed signal validation and electrical testing (voltage, current, signal integrity) using oscilloscopes, multimeters, and logic analyzers, achieving 99% test pass rate across 50+ device prototypes.
- **Firmware Integration:** Supported firmware integration and debugging for microcontroller-based telematics devices, improving sensor data processing accuracy by 15% and reducing power consumption by ~8%.
- **Data Analysis:** Analyzed telematics logs (location, speed, power status, fault codes) from 200+ test runs, identifying performance issues and supporting reliability improvements that reduced fault reports by 20%.
- **Hardware Development:** Assisted engineers with hardware bring-up, wiring, and PCB validation for 15 prototype boards, including connector verification and power supply testing, ensuring compliance with design specifications.
- **Documentation:** Prepared 6+ technical reports and test summaries, including procedures, results, and recommendations, supporting cross-functional team review and product validation.

PROJECT EXPERIENCE

Miniature Ultrasound Transducer for Wearable Medical Imaging 2026

- **FPGA Design & Signal Processing:** Developed a 16-channel preprocessing pipeline handling 1 MS/s per channel, including synchronized acquisition, filtering, envelope detection, and decimation, enabling real-time wireless transmission with <5 ms latency.
- **Control Interface Implementation:** Designed and implemented FPGA-AFE interfaces for programmable excitation timing, channel selection, VGA gain control, and Doppler support, improving signal quality by ~20% and supporting multiple scanning modes.
- **Modular Hardware Development:** Contributed to a scalable PCB architecture separating FPGA SoM, high-voltage pulser, and 16-channel AFE, reducing debugging time by 30% and supporting easy system upgrades for future 32-channel expansion.

FPGA SD-card Reader 2025

- **FPGA Development:** Implemented an SD-card host in SystemVerilog supporting SDv1, SDv2, SDHC cards, and FAT16/FAT32 file systems, enabling read/write operations on cards up to 64 GB.
- **Data Handling & File Access:** Built sector-level read logic and file-system parsing to locate and stream over 500 non-contiguous files, achieving continuous data throughput of ~2 MB/s with no data corruption.
- **Validation & Testing:** Performed full RTL simulation and on-board verification, reducing read/write errors and ensuring reliable operation across 5 different SD-card models.

Cardio Health Monitor 2024

- **Embedded System Development:** Implemented a real-time heart rate monitoring system in C, providing live visualization of heart rate variations from 50–180 BPM on an oscilloscope and health feedback on an LCD display.
- **Sensor Design & Integration:** Developed a PPG sensor with analog front-end (AFE) and ADC for amplified pulse detection, achieving ± 2 BPM measurement accuracy under varying heart rate conditions.
- **Testing & Verification:** Validated signal integrity and timing using laboratory instrumentation over 50+ test cycles, ensuring reliable readings and stable system performance.

TECHNICAL SKILLS

- **Programming / Software Skills:** C, C++, Python, Verilog/SystemVerilog, Vivado, ModelSim, Quartus, MATLAB, Power BI/Tableau, VBA
- **Embedded & Systems:** MCU and FPGA-based systems (STM32, ESP32, DE1-SoC, Xilinx), peripheral communication (CAN, UART, SPI, I2C), real-time systems, system-level debugging and validation
- **Laboratory & Testing:** Waveform generator, oscilloscope, multimeter, soldering, power supply